

ReviewMind

AI-Powered PR Reviewer That Learns Your Team's Coding Taste

PRODUCT
TYPE:

AI-Powered GitHub Agent

BUILDER:

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SUBMISSION:

Phase 1 Working MVP

GITHUB REPO:

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The Problem We Are Solving

Current software engineering workflows face severe code review friction.

1. Time-Consuming Reviews

Manual code reviews consume hours of developer time daily. Reviewers are forced to repeat simple syntax suggestions and style guidelines over and over, distracting them from architecture and business logic reviews.

2. Style Disconnects

Different teams have unique stylistic preferences. Standard static style guides are quickly forgotten, and code quality suffers when style expectations are only communicated verbally or through ad-hoc PR comments.

3. Static Linters Do Not Learn

Traditional static analysis tools check rigid rules. They do not adapt to your decisions. If a team intentionally overrides a pattern, static tools continue flags and noise indefinitely, leading to alert fatigue.

Proposed Solution & User Journey

ReviewMind combines AI-powered code reviews with adaptive machine learning.

ReviewMind: The Adaptive Reviewer

- **Context-Aware Reviews:** Analyzes changes, naming, error handling, and performance using *gpt-4o-mini*.
- **Adaptive Memory:** Learns from merged code/developer actions and stores decisions in SQLite.
- **Automated Fix PRs:** Automatically opens secondary pull requests with code modifications, reducing developer friction.
- **Dynamic Style Writing:** Compiles accepted reviews into a live style guide (**TEAM_STYLE.md**) committed back to the repo.

The 4-Step Developer Journey

Step 1: Code Pushed

Developer opens/updates a PR. GitHub triggers the ReviewMind Flask Webhook.

Step 2: AI Analyzes

ReviewMind reviews the diff, comments suggestions on GitHub, and opens a Fix PR.

Step 3: Feedback Loops

Dev merges the Fix PR (Auto-Accept) or comments *reviewmind reject* to train the model.

Step 4: Style Evolution

Once 20 feedbacks are gathered, ReviewMind writes and commits the *TEAM_STYLE.md* guide.

Tech Stack, Audience & Phase 1 Metrics

A lightweight, production-ready design built with robust integrations.

Tools & Technologies Used

- **Backend Framework:** Python & Flask (manages webhooks, API logic, and HTML template endpoints)
- **Database Storage:** SQLite (manages code diffs, reviews, learning patterns, and snapshots)
- **APIs & Libraries:** *PyGithub* (fetches diffs, creates fix branches, posts comments) and *OpenAI API* (generates reviews)
- **Dashboard Frontend:** Vanilla HTML5, CSS Variables, and *Chart.js* (displays acceptance graphs and style trends)

Target Audience & MVP Status

Target Audience: Software developers, engineering teams, and tech leaders striving to maintain clean, standardized code without manual review overhead.

Phase 1 MVP Production Run Results:

- **Total Reviews:** 22 PRs reviewed
- **Feedback Database Records:** 22 learning rows stored
- **Acceptance Rate:** 68% (based on real human interaction)
- **Fix PRs Opened / Merged:** 15 / 15
- **TEAM_STYLE.md:** Successfully generated and committed